

HDOC Day-26 - Complete Solution

C program using a menu, switch statement and nested loops to print right-aligned patterns. Each activity includes an algorithm and test case.

1. Right-aligned increasing triangle

Algorithm:

1. Start.
2. Read number of rows n.
3. For i = 1 to n:
 - a. Print (n-i) leading spaces.
 - b. Print i asterisks.
 - c. Move to the next line.
4. Stop.

Example for n = 4:

```
      *
     * *
    * * *
   * * * *
```

2. Right-aligned increasing number triangle

Algorithm:

1. Start.
2. Read number of rows n.
3. For i = 1 to n:
 - a. Print (n-i) leading spaces.
 - b. Print natural numbers from 1 to i.
 - c. Move to the next line.
4. Stop.

Example for n = 4:

```
      1
     1 2
    1 2 3
   1 2 3 4
```

3. Right-aligned decreasing triangle

Algorithm:

1. Start.
2. Read number of rows n.
3. For i = n down to 1:
 - a. Print (n-i) leading spaces.
 - b. Print i asterisks.
 - c. Move to the next line.
4. Stop.

Example for n = 4:

```
* * * *
 * * *
  * *
   *
```

C Program

```
#include <stdio.h>

void increasingStar(int n)
{
    int i, j;
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}

void increasingNumber(int n)
{
    int i, j;
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("%d ", j);
        printf("\n");
    }
}

void decreasingStar(int n)
{
    int i, j;
    for (i = n; i >= 1; i--) {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}

int main()
{
    int choice, n;

    printf("MENU\n");
    printf("1. Right-aligned increasing triangle\n");
    printf("2. Right-aligned increasing number triangle\n");
    printf("3. Right-aligned decreasing triangle\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter number of rows: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Number of rows must be positive.\n");
        return 0;
    }

    switch (choice) {
        case 1:
            increasingStar(n);
            break;
        case 2:
            increasingNumber(n);
            break;
        case 3:
            decreasingStar(n);
            break;
        default:
            printf("Invalid choice.\n");
    }

    return 0;
}
```

Test-Case Table

Test	Choice	n	Expected result
1	1	4	Right-aligned increasing star triangle with 4 rows
2	2	4	Right-aligned number triangle: rows end with 1, 2, 3, 4
3	3	4	Right-aligned decreasing star triangle from 4 stars to 1
4	4	4	Invalid choice
5	1	0	Number of rows must be positive

Result: The menu-driven C program successfully prints the selected right-aligned pattern using a switch statement and nested loops without arrays.